ASME STUDENT DESIGN COMPETITION

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THE ASME ENGINEERING GAMES

P O D I U M

Five Disciplines. One Platform.

OFFICIAL RULES

Calendar Year 2027 • Stage 1 of The ASME Engineering Games

Version 1.0

Important Notices

Authority of officials. Nothing in this document overrides the judgment of event Officials or any venue, local, or institutional safety requirement. Officials may stop any Robot or activity at any time. The Head Judge has final authority on site.

Rule questions. Rules are clarified through the published Q&A on the ASME SDC website (Section 4). Published answers are official and apply equally to every team.

Safety governs. Safety takes priority over performance and competition at all times.

Table of Contents

Important Notices 1
Table of Contents 2
1. Introduction and Intent 3
1.1 The Student Design Competition 4
1.2 A two-season program: The ASME Engineering Games 4
1.3 Design intent for PODIUM 4
2. The Spirit of the Games 5
3. Definitions 5
3.1 General terms 5
3.2 Officials 6
3.3 Discipline terms 7
4. Official Communications and Rules Q&A 8
5. Eligibility and Teams 8
6. Competition Format, Schedule, and Day-of Operations 8
6.1 The two days 8
6.2 Check-in and scheduling 9
6.3 Run timing and attempts 9
6.4 Field acceptance and replacement runs 9
6.5 Confirming scores 9
7. The Robot 10
7.1 One robot, size, and weight 10
7.2 Materials and design freedom 10
7.3 Power, energy, and control 10
7.4 Markings, required fittings, and prohibited features 11
8. Inspection 11
8.1 Inspection checklist 12
9. Safety 12
10. Conduct, Penalties, and Gameplay 13
10.1 Penalty framework 13
11. Fields, Game Objects, and How a Run Works 14
11.1 The Fields 14
11.2 Game Objects 14
11.3 How a Run works 15
11.4 The Crew 15
12. The Five Disciplines 15

12.1 The Sprint 16
Field and layout 16
How it is played 16
Scoring 17
12.2 The Lift 17
Field and layout 17
How it is played 17
Scoring 18
12.3 The Cycle 18
Field and layout 18
How it is played 18
Scoring 19
12.4 The Marksman 19
Field and layout 19
How it is played 19
Scoring 20
12.5 The Ascent 20
Field and layout 20
How it is played 20
Scoring 20
13. Scoring, Rankings, and Tiebreakers 21
14. The Playoff: Eliminations 21
15. Awards and Recognition 22
15.1 Two kinds of champion 22
15.2 Performance recognition 22
15.3 Judged recognition 23
16. The Engineering Design Report and Judging 23
16.1 Design report rubric 24
17. Looking Ahead: the CY2028 Season (Stage 2) 24

1. Introduction and Intent

This section introduces the competition for students, faculty, and advisors. The binding rules begin in Section 3.

1.1 The Student Design Competition

In the ASME Student Design Competition (SDC), teams of undergraduate ASME Student Members take an engineering problem from a blank page to a working Robot that competes on a Field. Each season is built around one design challenge. The competition welcomes every engineering discipline and rewards the full arc of real engineering: understanding requirements, generating concepts, building under constraints, testing, improving, and performing under pressure.

1.2 A two-season program: The ASME Engineering Games

The CY2027 and CY2028 seasons form a two-stage program, The ASME Engineering Games. This season, PODIUM, is Stage 1. The two stages are designed to build on each other.

One platform. Both seasons are contested by the same class of team-built Robot. The Robot size, weight, and safety rules are expected to remain the same next season, with at most minor clarifications, so a platform built well this year remains legal next year. Teams are encouraged to build a modular, upgradeable Robot rather than starting over.

Five Disciplines. Each season presents five scored Disciplines, each rewarding a different engineering strength. In Stage 2 some Disciplines will return, some will return in a harder form, and some will be new, with at least one returning. Stage 2 will be written so a brand new team is fully competitive.

Because students graduate and rosters change, nothing in this program assumes a team or student competes in both seasons.

1.3 Design intent for PODIUM

PODIUM is a multi-event contest, closer to a track and Field meet than a single game. The five Disciplines reward power, accuracy, speed and agility, throughput, and climbing. No Robot can excel at all five, so teams choose where to specialize. Every Discipline has a low entry point and a high ceiling, and the harder Disciplines are worth more points. The overall standing sums a team's best result in every Discipline, and no single Discipline can be farmed for unlimited points, so a contending team must perform across all five. Scoring is objective and quick so an Official can total it at the Field in seconds; design quality and documentation are recognized through a separate awards program (Sections 15 and 16) and do not affect performance scores.

In short: The Sprint rewards speed and agility; The Lift, strength and power; The Cycle, throughput and handling; The Marksman, accuracy and autonomy; and The Ascent, climbing and autonomy. The five stations come together at an event as shown below (see Section 11 for the full-size figure and details).

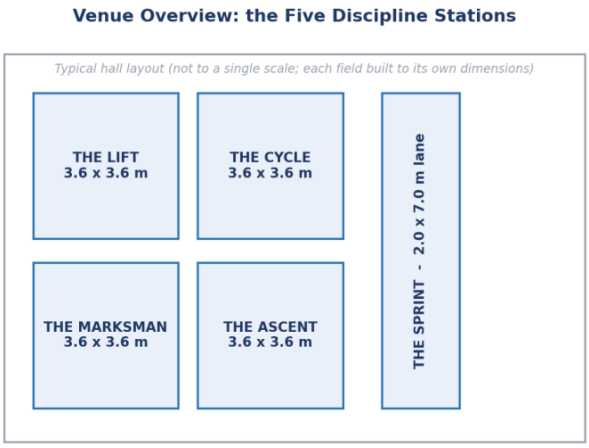


Figure 1 (reduced). Typical arrangement of the five Discipline stations in the venue.

2. The Spirit of the Games

Teams, mentors, volunteers, and officials are expected to uphold these principles. Officials may act on violations under Section 11.

Excellence. Pursue your best work and your best performance.

Ingenuity. Favor original, well-reasoned engineering. The SDC rewards teams that understand why their Robot works.

Respect. Treat competitors, volunteers, Judges, and equipment with respect. Disagree professionally and follow the decisions of Officials.

Safety. Safety comes before performance, always.

Sportsmanship. Compete hard, then help the team next to you. Rivals on the Field are teammates off the Field The playoff joins two teams whose strengths complement each other.

3. Definitions

Capitalized terms have the meanings below throughout these rules. Where an everyday word and a defined term could be confused, the defined term governs.

3.1 General terms

Term	Meaning
Robot	The single device a team designs, builds, and operates to compete.
Discipline	One of the five scored events: The Sprint, The Lift, The Cycle, The Marksman, and The Ascent.
Run	A single scored attempt by one Robot at one Discipline, consisting of an Autonomous Phase followed by a Crewed Phase. Every Run counts; there are no practice Runs.

Term	Meaning
Autonomous Phase	The first 30 seconds of a Run, during which the Robot acts only under its own onboard program or stored energy, with no human input.
Crewed Phase	The 2 minutes after the Autonomous Phase, during which the Crew operates the Robot with the Operator Console.
Crew	The up to four team members allowed at the Field for a Run. The Crew may change between Runs but must always be registered members of that team.
Operator Console	The controller and associated equipment the Crew uses to operate the Robot. It may be tethered or wireless. Teams supply their own; ASME does not provide a control system.
Station	The area where a single Discipline is set up and run, including its Field and the adjacent space where teams are received, queued, and scored.
Field	The bounded playing area of a Discipline, including its structures and Game Objects. The boundary is the vertical plane rising from the marked edge of the playing area.
Starting Zone	The marked area of a Field where the Robot is placed before a Run. The Robot must fit within a 40 cm cube here (Section 7).
Game Object	A game object used to score: Dumbbells (The Lift), Balls (The Cycle), and Bean Bags (The Marksman).
Discipline Score	The points a Robot earns in one Discipline on its Run, after that Discipline's formula and scoring limit are applied.
All-Around Score	The sum of a team's best Discipline Score in each of the five Disciplines.
Pair	Two teams joined to compete together in the playoff (Section 14).
Team Identifier	A visible label the team provides and attaches to its Robot showing its team name as registered. ASME does not provide identifiers.

3.2 Officials

Officials enforce these rules as written; only the Head Judge may interpret them. Judges evaluate engineering design reports only.

Role	Meaning
Official	Any volunteer or staff member ASME designates to run the competition.
Head Judge	The senior Official at a competition, with final authority on site for rulings, safety, and scheduling.

Role	Meaning
Scoring Official	The Official at a Station who starts Runs, totals the Discipline Score, and confirms it with the Crew.
Inspector	An Official who verifies that a Robot meets Sections 8 to 10 before it competes.
Judge	A volunteer who evaluates engineering design reports for the awards (Sections 15 and 16).

3.3 Discipline terms

Term	Meaning
Gate	In The Sprint, a 60 cm wide opening the Robot must pass through, in order, on alternating sides.
Hurdle	In The Sprint, a 5 cm tall barrier spanning the lane that the Robot should cross by driving over it.
Reset	In The Sprint, returning a stalled or off-course Robot to the last Gate it passed; each Reset carries a scoring penalty (Section 12.1).
Dumbbell	In The Lift, a coated weight with a center grip: Light (1 kg), Standard (2 kg), or Heavy (4 kg).
Tier	In The Lift, one of the three podium heights (20, 40, 60 cm), worth a multiplier of 1, 2, or 3 from low to high.
Ball	In The Cycle, a standard plastic ball-pit ball, about 65 mm across, hollow and soft.
Low Goal	In The Cycle, a floor-level pass-through goal, worth 2 points per Ball.
High Goal	In The Cycle, a raised pass-through goal at 1.0 m, worth 5 points per Ball.
Bean Bag	In The Marksman, a soft, round bag about 13 cm across that lands and stays where it comes to rest.
Target	In The Marksman, the flat ring target on the floor. A Bean Bag scores by the ring it rests in (more than half inside).
No-Cross Zone	In The Marksman, a line on the floor the Robot may not cross. Parts touching the floor stay behind it; an arm or launcher may reach over it in the air.
Loader	In The Marksman, one Crew member who hands Bean Bags to the Robot one at a time during the Crewed Phase.
Structure	In The Ascent, the vertical climbing wall the Robot ascends, fitted with rungs, holds, and bars. The Robot may grip and attach to it.
Height Band	In The Ascent, a scored height measured from the floor to the lowest

Term	Meaning
	point of the Robot (0.5, 1.0, 1.5 m).
Summit	In The Ascent, the top condition: the lowest point of the Robot held at 2.0 m or higher with nothing touching the floor.

4. Official Communications and Rules Q&A

4-1 Official channel. These rules, the Field drawings, the schedule, and every update are posted on the ASME SDC website. Teams are responsible for checking it regularly.

4-2 Questions. Teams submit rules questions through the official online Q&A form linked on the ASME SDC website and SDC Discord Channel. Answers of general interest are published so that every team sees the same information. Q&A updates will be provided every two weeks.

5. Eligibility and Teams

5-1 A team may have any number of members. Every member must be an undergraduate engineering student and a current ASME Student Member at the time of the competition. Any engineering discipline is welcome. On the date of the competition, the member must not yet have received a bachelor's or equivalent degree in engineering; students who have not completed their undergraduate coursework, including community-college and associate-degree students, are eligible.

5-2 Each student may contribute to only one team for the season and may not work on another team's Robot.

5-3 Each team competes with exactly one Robot. A school may enter more than one team, but each team has its own single Robot.

5-4 Each team designates one student Team Captain as its contact with Officials. The Captain must be a team member and an ASME Student Member.

5-5 Each team has at least one faculty or professional mentor of record, named at registration. Mentors advise; the Robot must be the students' work.

5-6 Teams register in advance through the website for the EFx they wish to attend, by the published deadlines. A student may participate at the event only after all required forms for that student are submitted.

6. Competition Format, Schedule, and Day-of Operations

6.1 The two days

- **Day 1, Qualification.** Check-in, Robot inspection (Section 8), and one Run at each of the five Disciplines. At the end of Day 1, All-Around Scores are posted and teams are ranked.

- **Day 2, Playoff and Awards.** Pair selection and the elimination playoff (Section 14), then the awards (Section 15).

6.2 Check-in and scheduling

6.2-1 Every team must complete check-in by the time posted on the EFx event schedule. If a team does not meet the check-in window it is disqualified from the competition. In exceptional circumstances the Head Judge may extend the check-in close for all teams at once, announced publicly; extensions are never granted to individual teams.

6.2-2 The schedule is built from the check-in list. After check-in closes, the list of present teams is randomized to assign each team an order through the Disciplines. The schedule is posted immediately after check-in closes. Teams must be checked in and ready to compete from the start of the day, since a team may be scheduled first.

6.2-3 A team that is not present and ready when called to a Discipline is disqualified from that Run. There is no grace period.

6.2-4 Inspection is integrated into the schedule: a team is inspected before its first Discipline (Section 8).

6.2-5 There are no practice Runs. Every Run counts.

6.3 Run timing and attempts

6.3-1 When called to a Discipline, a team has 60 seconds to place its Robot in the Starting Zone and be ready. If the team is not ready within 60 seconds, the Scoring Official may start the Run regardless; the team must place its Robot and begin immediately, and any time lost is not compensated. A Run lasts 2 minutes 30 seconds: a 30-second Autonomous Phase then a 2-minute Crewed Phase. The Station is then cleared for the next team.

6.3-2 Each team gets one Run per Discipline, which produces that team's Discipline Score. If time allows, an event may schedule additional Runs for all teams, in which case a team's best score in each Discipline counts.

6.4 Field acceptance and replacement runs

6.4-1 Before a Discipline's first Run, teams may view that Field from the viewing area designated by the Head Judge. Teams may not approach or touch a Field at any time. Any concern about a Field must be raised before that Discipline's first Run begins. Once that first Run has started, the Field is accepted as it is for the rest of the event, and Runs are not replayed for Field conditions.

6.4-2 A replacement Run is given only for a clear malfunction of scoring or timing equipment, a clear Official error, or proven outside wireless interference during a specific Run, at the Head Judge's discretion.

6.5 Confirming scores

6.5-1 The Scoring Official totals each Discipline Score at the Station right after the Run and shows it to the Crew, who confirm it before leaving. Once confirmed, or once the team

leaves the Station, the score is final and cannot be disputed. Each Station keeps the standard score sheet so scores can be totaled and shown within seconds.

7. The Robot

These rules define a legal Robot while preserving the open-design freedom of the SDC.

7.1 One robot, size, and weight

7.1-1 Each team competes with exactly one Robot for the season.

7.1-2 At the start of every Run, and at inspection, the Robot must fit within a 40 cm by 40 cm by 40 cm cube on its own, with nothing held back by hand. Each team provides its own rigid container no larger than 40 cm on a side that holds the Robot for sizing and between Runs. Rigid means the container is self-supporting and retains its shape and dimensions when empty; it must not collapse under the weight of its own sides when lifted by a corner, and flexible materials such as cardboard or fabric are not acceptable. At events where rigid materials are impractical, the Head Judge may approve a container of locally available material (for example cardboard).

7.1-3 After the start signal the Robot may expand, but never above 75 cm in any horizontal direction and never above 120 cm tall, and never unsafely.

7.1-4 The Robot must weigh no more than 15 kg as it competes, including onboard batteries and stored air. The Operator Console, any tether, tools, spare parts, and the team's laptop are not counted.

7.2 Materials and design freedom

7.2-1 Teams may use commercial off-the-shelf and custom-fabricated parts of any material, provided the Robot is safe.

7.3 Power, energy, and control

7.3-1 The Robot runs on electricity from onboard batteries, plus the pneumatics and stored mechanical energy allowed below. No internal-combustion engines, fuel, or combustion of any kind.

7.3-2 No single battery may exceed 24 V nominal, and the control system must run at 24 V nominal or below. Batteries must be commercially made, used within their ratings, undamaged, and securely mounted, with an easily reached main power switch or breaker. Any battery chemistry is allowed, but every team must store and charge its batteries in a fire-safe charging bag (Section 9).

7.3-3 Stored mechanical energy (springs, elastic, and the like) may be used. A stored-energy device may be pre-loaded before a Run, provided its potential energy can be fully restored by the Robot itself during the Run (for example, by re-winding a winch or re-compressing a spring). A device may not be pre-loaded in a way that stores more energy than the Robot can regenerate from its starting state.

7.3-4 No flight. Aerial propulsion of any kind is prohibited, whether tethered or not. A Robot may not lift itself off a surface using thrust, rotors, lighter-than-air methods, or similar.

7.3-5 Teams supply their own control system. The Robot uses one control link: either a tether or a single wireless link. Wireless control must use only license-free consumer bands (such as 2.4 GHz or 5 GHz) at legal power. The team is responsible for its link not interfering with the venue or another team, and for being able to disable the Robot on request. A team whose link interferes must resolve it or be disabled. If a tether is used, it must remain completely slack and trailing behind the Robot, and it may not be connected to any stationary object or person. It is not part of the Robot, is not counted for weight, and is not measured for height in The Ascent. Any tension observed pulling, lifting, or steadying the Robot is a violation and draws a Minor Penalty; wireless control is encouraged. The safety line an Official attaches for The Ascent (Rule 7.4-3) is not a tether under this rule. Autonomous control may use an onboard processor or the stored-energy methods of Rule 7.3-3.

7.3-6 Pneumatics may use commercial components. The maximum working pressure anywhere on the Robot is 60 psi, which is below the rating of common pneumatic parts. The Robot must carry a visible pressure gauge and a relief valve. Hydraulic systems, if used, must operate below 100 psi and use a non-flammable, food-grade hydraulic fluid.

7.4 Markings, required fittings, and prohibited features

7.4-1 The Robot clearly displays its Team Identifier and provides a flat area at least 3 cm by 3 cm where an Official attaches a colored marker (tape) to show its Pair during the playoff.

7.4-2 Prohibited: anything meant or likely to damage the Field, Game Objects, or another Robot; launches that exceed low, safe energy or that send Game Objects out of the Field; sharp or hazardous edges; lasers above Class 1; materials that shatter dangerously; mercury or hazardous chemicals; pyrotechnics or explosives; and any wireless device that interferes with event operations or another team.

7.4-3 Ascent anchor point. The Robot must include an accessible anchor point near its top (for example a loop or eyebolt) capable of supporting the Robot's full weight. Before each Run in The Ascent, an Official attaches the venue's safety line to this point with a carabiner (Section 12.5).

8. Inspection

8-1 Every Robot must pass inspection before its first Run. A Robot that has not passed may not compete. Inspection is performed at the team's first Discipline by the Officials there (Section 6); allow 5 to 10 minutes per team.

8-2 Inspection checks the items in Sections 7, 9, and 10. The checklist below is the one Inspectors use; teams should self-check against it first.

8-3 Repairs and re-inspection. Between Runs, teams may make minor repairs and adjustments at their team table, including replacing broken parts with equivalents and changing code on a laptop. Major modifications are not allowed: a major modification is any change to the Robot's fit, form, or function. Any change that could affect the Robot's compliance with Sections 7, 9, or 10, including its size, weight, or safety state, requires re-inspection before the Robot competes again. Teams are expected to maintain the spirit of the competition and not perform a complete redesign between Runs. Final determination of compliance rests with the Inspector; disputes are resolved by the Head Judge. The laptop is not part of the Robot and may not control or assist the Robot during a Run beyond the Operator Console.

8.1 Inspection checklist

Item	Rule(s)
Exactly one Robot for the team	Rule 7.1-1
Fits within the team's 40 x 40 x 40 cm cube on its own	Rule 7.1-2
Expansion within limits (75 cm wide, 120 cm tall)	Rule 7.1-3
Weighs 15 kg or less as competed	Rule 7.1-4
Electric power only; no combustion	Rule 7.3-1
Battery 24 V nominal or less; secure; undamaged; main switch	Rule 7.3-2
Fire-safe charging bag present	Rule 7.3-2, Rule 9-2
Stored-energy pre-load can be restored by the Robot itself	Rule 7.3-3
No flight or aerial propulsion	Rule 7.3-4
One control link on a license-free band; can be disabled; tether slack if used	Rule 7.3-5
Pneumatics: visible gauge reads 60 psi or less when charged; relief valve present	Rule 7.3-6
Ascent anchor point present, accessible, and rated for the Robot's full weight	Rule 7.4-3
Team identifier shown; flat area for Pair marker	Rule 7.4-1
No prohibited or hazardous features; safe edges	Rule 7.4-2
Team has its own eye protection marked Z87 or EN166 or equivalent marking on the frame or lens	Rule 9-2

9. Safety

9-1 Safety comes before performance and competition. Officials may stop any Robot, Run, or activity at any time.

9-2 Eye protection. Everyone must bring and wear their own safety eyewear in the team and competition areas whenever Robots are worked on or operated. Eyewear must meet at least the ANSI/ISEA Z87.1-2020 or EN 166 impact standard. Inspectors will check for the Z87.1, Z87+, EN166, or equivalent marking on the frame or lens. ASME does not provide eye protection.

9-3 Battery safety. Batteries are stored and charged in a fire-safe charging bag, handled per manufacturer guidance, and any damaged or swollen battery is removed from service and immediately reported to the Head Judge.

9-4 Each Robot can be brought to a safe stop quickly with its main switch or control link, and a Crew must be able to disable its Robot when an Official asks.

9-5 A Robot may be operated only on a Field during its scheduled Run, with Official approval. At the team table a Robot may be powered for repair and inspection but not driven. Stored energy and pressure must be kept safe during handling and transport.

10. Conduct, Penalties, and Gameplay

These rules keep play safe, fair, and professional.

10-1 Professional conduct is expected at all times.

10-2 Teams and Robots may not deliberately damage the Field, Game Objects, or another Robot.

10-3 A Robot may not interfere with another team's Run or scoring. No defense of any kind is permitted, including during playoff matches (Section 14).

10-4 A Robot may attach to or grip the Field structure when a Discipline calls for it, such as climbing in The Ascent. A Robot may never grapple, attach to, tip, or entangle another Robot.

10-5 A Robot that leaves its Field during a Run may be disabled by Officials for the rest of that Run.

10-6 Once a Run starts, the Crew may not contact the Robot except when an Official allows it, such as a safety stop. A safety stop ends the Run.

10-7 Designing a strategy whose only purpose is to force an opponent to break a rule is not allowed, and penalties caused that way are not charged to the targeted team.

10.1 Penalty framework

Officials apply the lightest penalty that fits, escalating as needed.

Level	Effect	Typical triggers
Warning	A verbal caution with no effect on score.	A minor delay, a Crew member briefly out of position, a first small infraction.
Minor Penalty	Minus 10 points from that Run's Discipline Score, or the affected	Brief unintended Field contact, exceeding the setup time limit, a single

Level	Effect	Typical triggers
	scoring is removed.	mishandled Game Object.
Voided Run	The Run scores zero.	Unsafe operation, significant interference with the Field or another team, contacting the Robot during a Run when not allowed, leaving the Field.
Formal Warning	A formal warning recorded for the event; a second Formal Warning becomes a Match Disqualification.	Repeated minor penalties or unsporting behavior.
Match Disqualification	Disqualification from the current Run or match, scoring zero.	A serious safety or conduct violation, or a second Formal Warning.
Event Disqualification	Removal from the Discipline or the event.	Egregious or repeated safety, cheating, or unsporting violations, at the Head Judge's discretion.

10.1-1 Officials make the final call at the Field. Concerns are raised respectfully and before a score is confirmed (Section 6). Video replay is not used.

11. Fields, Game Objects, and How a Run Works

11.1 The Fields

Each Discipline is set up at its own Station, sized for a typical gym or competition hall and built the same way at every event. The Lift, The Cycle, The Marksman, and The Ascent each use a low-walled Field about 3.6 m by 3.6 m. The Sprint uses its own lane, 2.0 m by 7.0 m, which fits alongside the others. Each Discipline in Section 12 includes a dimensioned diagram and its key dimensions; those diagrams and dimensions are the official layout for play.

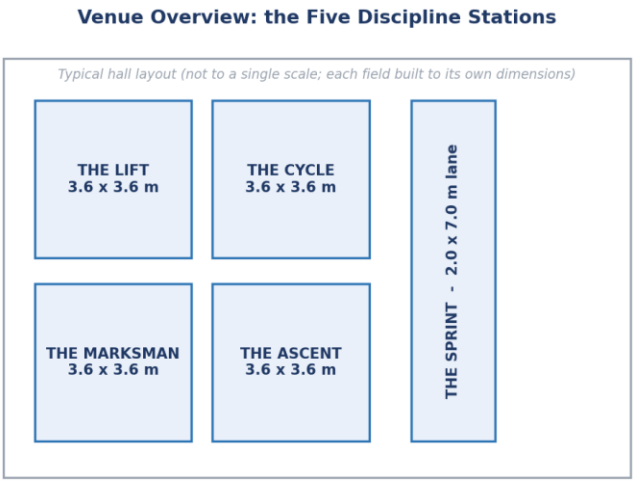


Figure 1. Typical arrangement of the five Discipline stations in the venue.

11.2 Game Objects

Game Objects are common, inexpensive, globally available items, chosen to be safe to handle and to be struck by a Robot. A shared supply is provided at each Station.

Element	Used in	Specification
Dumbbell	The Lift	A coated weight with a central hand grip, in three masses: Light 1 kg, Standard 2 kg, Heavy 4 kg, about 15 to 20 cm across, color coded by mass. (Standard vinyl- or rubber-coated fitness weights.)
Ball	The Cycle	A standard plastic ball-pit play ball, about 65 mm in diameter, hollow and soft, a few grams each.
Bean Bag	The Marksman	A round soft bean bag about 13 cm across and roughly 150 to 200 g, that lands and stays where it comes to rest. (Standard physical-education bean bags.)

11.3 How a Run works

11.3-1 Every Run is 2 minutes 30 seconds: a 30-second Autonomous Phase then a 2-minute Crewed Phase. The Sprint and The Ascent use this same window; The Sprint is scored on its finish time within the window, and The Ascent is scored at the final buzzer.

11.3-2 The Robot is brought to the Station inside its container. Before a Run, the Crew removes the Robot from the container and places it in the Discipline's Starting Zone on its own, within the 40 cm starting size, then steps back. At the start signal, one Crew member may press a single start control on the Robot or Operator Console to begin the Autonomous Phase, then must step away. A Robot that plans no Autonomous Phase action simply remains still until the Crewed Phase.

11.3-3 During the Autonomous Phase no one may touch the Robot or the Operator Console or send any input after the start control. The Robot runs on its own program or stored energy. Breaking this voids any Autonomous Phase bonus and may draw a penalty.

11.3-4 When the Crewed Phase begins, the Crew operates the Robot through the Operator Console.

11.4 The Crew

11.4-1 For each Run a team may bring up to four Crew members to the Field. The Crew may change between Runs but must always be registered members of that team.

11.4-2 Only Crew members may operate the Operator Console, handle the Robot when permitted, or speak with Officials at the Field during the Run. Other team members are welcome to cheer and call out from outside the Field; that spirit is encouraged. Mentors and other non-team adults may not operate the Robot or coach during a Run.

12. The Five Disciplines

Each Discipline below gives its objective, its Field and layout with dimensions, how it is played, and how it is scored. Disciplines are listed from most accessible to most demanding, which is also the order of how many points each is worth. Each Discipline has a fixed maximum, so no single one can be farmed for unlimited points.

Discipline	Inspired by	Engineering focus	Max points
12.1 The Sprint	Sprint and show jumping	Speed and agility	100
12.2 The Lift	Weightlifting	Strength and power	120
12.3 The Cycle	Lacrosse and flag football	Throughput and handling	140
12.4 The Marksman	Archery	Accuracy and autonomy	160
12.5 The Ascent	Sport climbing	Climbing and autonomy	180

12.1 The Sprint

Objective. Cross a course of Gates and Hurdles as fast as possible. Worth up to 100 points.

Field and layout

A lane 2.0 m wide and 7.0 m long, with a start line and a finish line 7.0 m apart. Four Gates, each a 60 cm wide opening, are spaced about every 1.4 m along the lane on alternating sides, forming a slalom. Two Hurdles, each 60 cm wide and 5 cm tall, are placed on alternating sides of the lane rather than spanning it, so a Robot may either cross over a Hurdle or pass it on the open side. The Starting Zone is at the start line.

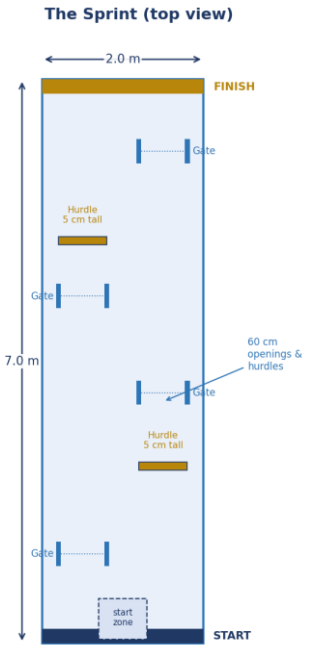


Figure 2. The Sprint lane (top view).

How it is played

The clock runs from the start signal until the Robot fully crosses the finish line, up to the 2 minute 30 second Run. The four Gates must be taken in order, on alternating sides, and both Hurdles should be crossed by driving over them. Missing a Gate, going around a Hurdle, and any Reset each add a penalty, as below. If the Robot passes through the first Gate on its own during the Autonomous Phase, the team earns a 10 point bonus. A Robot that does not cross the finish line within the Run scores 0 for The Sprint.

Scoring

Start at 100 points. Subtract 1 point for each second of finish time, and subtract the penalties below. Add the Autonomous Phase bonus if earned. The score is never below 0 and never above 100.

Item	Points
Each second of finish time	minus 1
Each Gate missed	minus 15
Each Hurdle gone around instead of over	minus 10
Each Reset	minus 15
First Gate passed in the Autonomous Phase	plus 10

12.2 The Lift

Objective. Place calibrated Dumbbells onto a three-tier podium. Heavier Dumbbells higher up score more. Worth up to 120 points.

Field and layout

Within the 3.6 m by 3.6 m Field, a podium with three flat tiers at 20 cm, 40 cm, and 60 cm stands against the far edge. Each tier has a top surface about 40 cm by 40 cm. A loading zone holding the Dumbbell supply (at least six of each mass) sits 1.0 m in front of the podium. The Starting Zone is behind the loading zone.

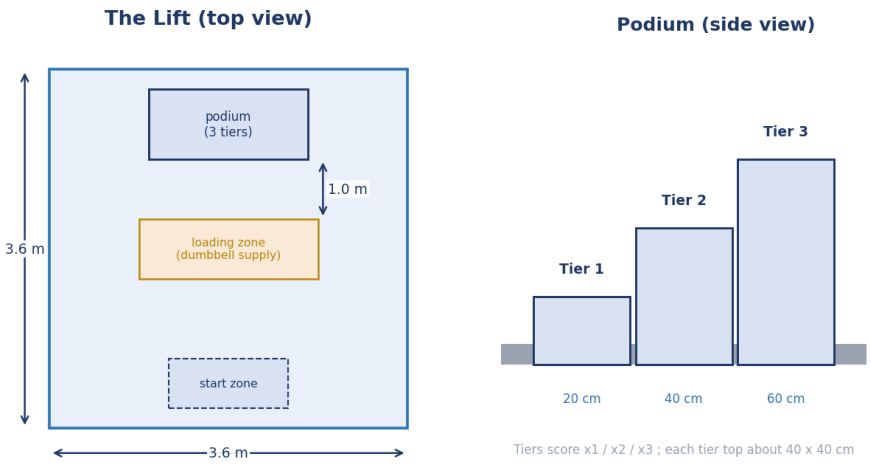


Figure 3. The Lift: Field layout (left) and podium tiers (right).

How it is played

The Robot carries Dumbbells from the loading zone and places them on the podium. A Dumbbell scores if, at the final buzzer, it is resting on a tier and supported by the podium rather than by the Robot. Dumbbells may be stacked; every Dumbbell in a stable stack scores, and the whole stack is counted at the tier its bottom Dumbbell rests on. A stack must be stable: if any Dumbbell in a stack would fall were the Robot not present, the entire stack does not score. Dumbbells placed during the Autonomous Phase are worth 1.5 times their value.

Scoring

Read each scored Dumbbell from the table, then multiply by 1.5 if it was placed during the Autonomous Phase. The maximum is 120. The final score will not exceed 120 points.

Dumbbell	Tier 1 (x1)	Tier 2 (x2)	Tier 3 (x3)
Light (1 kg)	5	10	15
Standard (2 kg)	10	20	30
Heavy (4 kg)	20	40	60

12.3 The Cycle

Objective. Collect Balls and score them through goals. This is the basis of the playoff. Worth up to 140 points.

Field and layout

Within the 3.6 m by 3.6 m Field, a Low Goal with a 40 cm wide floor-level opening and a High Goal with a 30 cm opening centered at 1.0 m stand on the far edge. A Ball supply sits at a loading area about 1.8 m from the goals. The Starting Zone is at the loading area. Both goals are pass-through: a Ball counts when it passes fully through the opening.

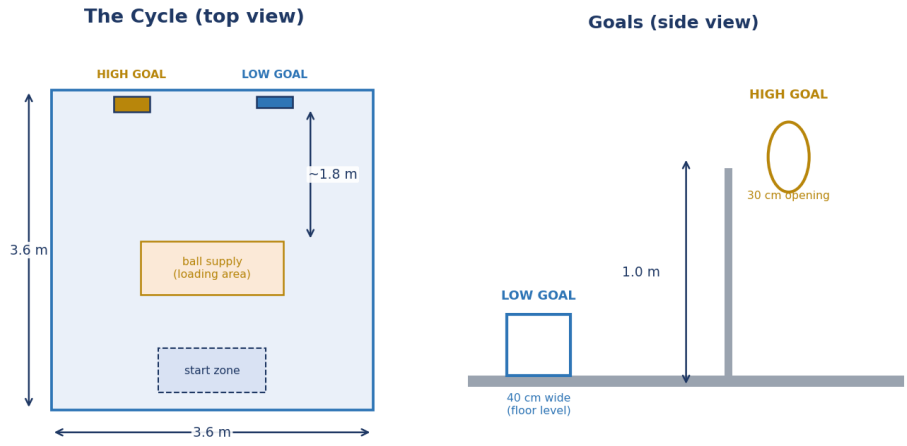


Figure 4. The Cycle: Field layout (left) and goals (right).

How it is played

The Robot collects Balls and scores them through the goals. A Ball counts when it passes fully through a goal opening. There is no limit on how many a Robot may score, though the maximum caps the points. A Robot may control at most three Balls at a time; deliberately

herding or pushing more than three draws a Minor Penalty. Balls scored during the Autonomous Phase count double.

Scoring

Score = (Low Goal balls x 2) + (High Goal balls x 5), with Autonomous Phase balls doubled. The maximum is 140. The final score will not exceed 140 points.

12.4 The Marksman

Objective. Land Bean Bags on a ringed Target as accurately as possible, with a Loader feeding one Bean Bag at a time. Worth up to 160 points.

Field and layout

Within the 3.6 m by 3.6 m Field, a flat ring Target, 60 cm across with five concentric rings, lies on the floor with its center 2.5 m from the No-Cross Zone line. This distance is too far to place a Bean Bag by reaching across, so Bean Bags must be launched, tossed, or slid. The rings score, from the bullseye outward, 20, 16, 12, 8, and 4 points. A loading station holding the eight Bean Bags sits behind the No-Cross Zone, where the Starting Zone is.

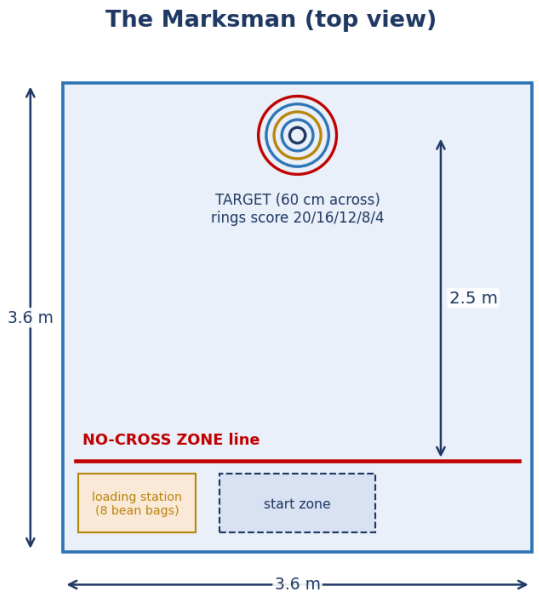


Figure 5. The Marksman (top view).

How it is played

The Robot delivers Bean Bags onto the Target from behind the No-Cross Zone. Parts of the Robot touching the floor must stay behind the line; an arm or launcher may reach over it in the air. To reward driving, the Robot may start a Run holding at most one Bean Bag, and during the Crewed Phase the Loader hands Bean Bags to the Robot one at a time. During the Autonomous Phase no one may touch the Robot or Bean Bags; the Robot may, on its own, launch the Bean Bag it started with and also collect additional Bean Bags from the loading station and launch those, all without human help. As a stricter, discipline-specific exception to Rule 7.3-3, any launching mechanism must start each Run in its unloaded state; pre-tensioned

launchers are not allowed in this Discipline. A Bean Bag scored during the Autonomous Phase counts double.

Scoring

Score = the sum of the ring values of all scored Bean Bags, with the Autonomous Phase Bean Bag doubled. The maximum is 160. The final score will not exceed 160 points.

12.5 The Ascent

Objective. Climb the wall. The higher the Robot is holding at the buzzer, the more it scores, and climbing on its own is worth much more. Worth up to 180 points.

Field and layout

Within the 3.6 m by 3.6 m Field, a vertical climbing wall at least 2.2 m tall stands against the far edge, fitted with rungs, holds, and bars, with a padded base. Three Height Bands are marked at 0.5 m, 1.0 m, and 1.5 m, measured from the floor to the lowest point of the Robot. The Summit is reached when the lowest point of the Robot is held at 2.0 m or higher with nothing touching the floor. The Starting Zone is on the floor in front of the wall.

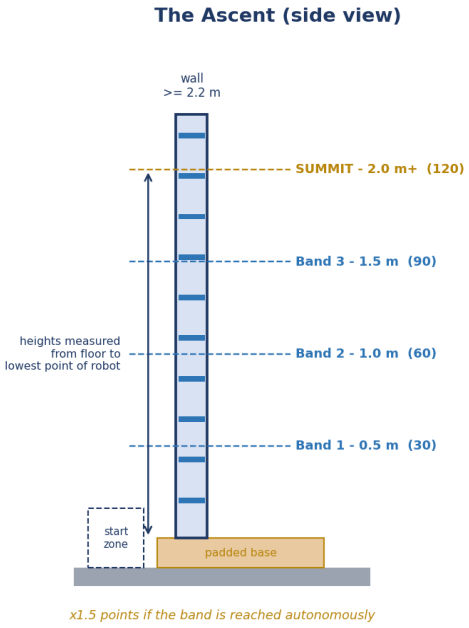


Figure 6. The Ascent wall (side view).

How it is played

Before each Run, an Official clips the venue's safety line to the Robot's anchor point (Rule 7.4-3) with a carabiner; the line stays slack, is not part of the Robot, is not measured for height, and may not assist the climb. The Robot climbs the wall and must be supported by it, not the floor. Only the highest Band it is stably holding at the final buzzer counts, and a Robot touching the floor at the buzzer scores 0. Climbing is the whole task; the Robot carries nothing up. If the highest Band held at the buzzer was first reached during the Autonomous Phase, the Ascent score is multiplied by 1.5.

Scoring

Score by the highest Band held at the buzzer, times 1.5 if that height was reached during the Autonomous Phase. The maximum is 180. The final score will not exceed 180 points.

Highest held at buzzer	Reached in the Crewed Phase	Reached in the Autonomous Phase (x1.5)
Band 1 (0.5 m)	30	45
Band 2 (1.0 m)	60	90
Band 3 (1.5 m)	90	135
Summit (2.0 m or higher)	120	180

13. Scoring, Rankings, and Tiebreakers

13-1 Each Discipline produces a Discipline Score using its own formula (Section 12), up to that Discipline's maximum: The Sprint 100, The Lift 120, The Cycle 140, The Marksman 160, The Ascent 180. No Discipline Score may exceed its Discipline's maximum; the maximum is applied after all bonuses and multipliers.

13-2 A team's All-Around Score is the sum of its best Discipline Score in each of the five Disciplines, up to 700. Teams are ranked by All-Around Score for the Day 1 standings and the playoff seeding.

13-3 Tiebreakers, in order: (1) the team with the higher score in its best Discipline wins; if still tied, compare second-best Discipline Scores, and continue down the list; (2) if still tied, the faster best finish time in The Sprint; (3) if still tied, a coin toss by the Head Judge.

13-4 Scores are posted as the day goes and confirmed at the Station when set (Section 6). The engineering design report and design quality do not affect the All-Around Score, Discipline recognition, or the playoff; they are recognized only through the judged awards (Sections 15 and 16).

14. The Playoff: Eliminations

Day 2 closes the event with a team-based playoff that rewards versatility and cooperation while keeping scoring as simple as addition.

14-1 Pair count. A set number of two-team Pairs contest the playoff, scaled to the number of teams at the event (for example, four to eight Pairs at most events). The number is announced before selection.

14-2 Selection comes first. The highest-ranked teams by All-Around Score pick the Pairs. In rank order, each of these teams invites one other qualified team to join its Pair; an invited team that declines stays available and may be invited by another team. All Pairs are formed this way before any elimination match is played, so every picking team draws from the same pool and no team is left to be paired by default.

14-3 Bracket and byes. The completed Pairs play a seeded single-elimination bracket, seeded by the rank of each Pair's higher-seeded team. If the number of Pairs is not a power of two, the top-seeded Pairs receive first-round byes. Because all Pairs are already formed before the bracket begins, a bye affects only the bracket, not selection.

14-4 Featured Disciplines. Each playoff match is contested in The Cycle plus one Featured Discipline. The Featured Discipline for each round is announced at the start of the playoff, after qualification, so teams and Judges know the full slate before selection. By default the semifinals feature The Sprint, the final features The Ascent, and the third-place match features The Marksman.

14-5 Match score. In a match, both partner Robots of a Pair each take one Run in each of the two featured Disciplines, and the Pair match score is the sum of those four Discipline Scores. At events with many teams, to save time the Head Judge may instead have each Pair run only one Robot in each featured Discipline, with the Pair choosing which of its Robots competes in which Discipline; the match score is then the sum of those two Discipline Scores.

Either way, the higher total wins and advances.

14-6 Running matches. Matches may run head to head on mirrored Fields, or one Pair at a time on a single Field, depending on the venue. No defense of any kind is permitted in any playoff match: any attempt to block, impede, or contact the opposing Robot or its Game Objects draws a penalty under Section 10.

14-7 Match ties. A tied match is broken the same way as Rule 13-3: by the higher single Discipline Score among the match's scores, then the next highest, then a coin toss by the Head Judge.

14-8 Titles. The winner of the final is the PODIUM Champion, the Gold Pair; the final's loser is the Silver Pair; the third-place match winner is the Bronze Pair.

15. Awards and Recognition

PODIUM recognizes both performance and engineering.

15.1 Two kinds of champion

Two top performance honors exist because they reward different things. The All-Around honor goes to consistent, broad performance across all five Disciplines during qualification, the best all-rounder. The PODIUM Champion is the winner of the Day-2 playoff, the best in head-to-head elimination. A team may aim for either or both.

15.2 Performance recognition

Recognition	Basis
PODIUM Champion (Gold, Silver, Bronze Pair)	The playoff result (Section 14). The headline title of the event. Recognition: Gold $1,000 ($750 to the higher-seeded

Recognition	Basis
	team, $250 to its ally); Silver $500 ($375 / $125); Bronze $250 ($190 / $60). One trophy per podium Pair, kept by the higher-seeded team.
All-Around	The highest teams by All-Around Score, the best all-rounders of qualification. Recognition: Trophy.
Discipline recognition	The top-scoring team in each of the five Disciplines. A team may earn more than one. Recognition: One trophy per Discipline.

15.3 Judged recognition

Judged recognition is based on the engineering design report and its rubric (Section 16).

Recognition	Basis
The Laurel Award	The top overall team: strong performance, an outstanding design report, and exemplary conduct. The best complete program at the event. Recognition: Certificate.
The Blueprint Award	The clearest, most rigorous engineering documentation and design process, judged from the design report. Recognition: Certificate.
The Ingenuity Award	The most original or clever engineering solution on a Robot. Recognition: Certificate.
The Spirit of the Games Award	Professional conduct and camaraderie: a team that competes hard and lifts up others. Recognition: Certificate.
The Rising Star Award	The strongest rookie team. A rookie team is one in which at least 51 percent of members are first or second-year students, verified from the year-of-study information collected at registration. If no eligible rookie team competes at an event, this award is not presented. Recognition: Certificate.

16. The Engineering Design Report and Judging

The engineering design report shows how a team's Robot was designed, why decisions were made, how it was tested, and how it improved. It is the basis for the judged awards and does not affect the performance recognition.

16-1 Each team keeps an engineering design report through the season covering problem analysis, concept generation and selection, design decisions and trade-offs, analysis and calculations, testing and iteration, and team and project management.

16-2 Submitting a design report is optional: a team that does not submit still competes normally but is not eligible for the judged recognition in Section 15.3. Teams that want to be considered submit their design report through the official online form by the posted deadline, 30 days before the event's first competition day; late submissions are not accepted. The submission is a single PDF of 10 to 30 pages (a cover page and a references section or appendix are not counted toward the limit); pages beyond 30 are not reviewed. Design reports are judged from the submission using the rubric below.

16-3 The Blueprint Award rewards the quality of the engineering documentation and design process shown in the design report, not Robot performance.

16-4 Judged-award ties. When design reports tie on total rubric score, the award goes to the higher score in Engineering analysis, then in Design process and decisions, and if still tied, to the Judges' panel decision.

16.1 Design report rubric

Each category is scored from 0 to 4 against the anchors below; levels 1 and 3 fall between them. The five categories total 20 points.

Category	0 - Absent	2 - Developing	4 - Exemplary
Design process and decisions	No process shown.	Some concepts and choices, lightly justified.	Clear path from requirements to concepts to a justified design, with trade-offs.
Engineering analysis	No analysis.	Basic calculations in places.	Sound analysis of loads, motion, power, and tolerances driving real decisions.
Build, testing, and iteration	No testing shown.	Some testing, limited change.	Repeated test-and-improve cycles with evidence of what changed and why.
Documentation quality	Hard to follow.	Readable but uneven.	Clear, well organized, and easy for a newcomer to follow.
Project and team management	None shown.	Some planning.	Clear roles, schedule, and resource tracking carried through the season.

17. Looking Ahead: the CY2028 Season (Stage 2)

This is shared so teams can make smart design choices now. It is general on purpose; Stage 2 is fully specified in its own rules and written so teams new in CY2028 are completely competitive.

- **Same platform and Robot rules.** Stage 2 is contested by the same class of Robot, and the size, weight, safety rules, and inspection are expected to stay the same, with at most minor clarifications. A modular, upgradeable Robot built now will be legal next year.
- **Five Disciplines, mixed.** Expect five scored Disciplines: some returning, some returning harder, and some new, with at least one returning so a subsystem built now stays useful.
- **Rising challenge.** Expect overall difficulty and the points tied to Autonomous Phase performance to increase.
- **Teamwork and endurance.** Expect more challenges that reward working together and performing consistently over time.
- **Defensive play.** Head-to-head defensive play is being considered for Stage 2, informed by how the CY2027 season plays out.
- **Iteration recognized.** A modest recognition is planned for teams that clearly evolve their platform from Stage 1 to Stage 2, shown through the design report, without giving returning teams a scoring edge over new teams.